

Habib University

Course Code: EE252/CE251- Spring 2025

Course Title: Signals and Systems

Instructor's Name: Tariq Mumtaz

Examination: Midterm-I

Total Marks: 60 (15% of Course credits)

Date: 20th Feb , 2025

Duration: 75 minutes

	Course Learning Outcomes	Cognitive domain Learning Domain Level
CLO 1	To represent, model, and analyze continuous and discrete signals and systems in the time domain.	LDL = 3 Application PLO=01 Engineering Knowledge

Attempt all questions, your answer must include neat and properly labeled sketches (where applicable), a logical flow of the solutions, and detailed steps of your work.

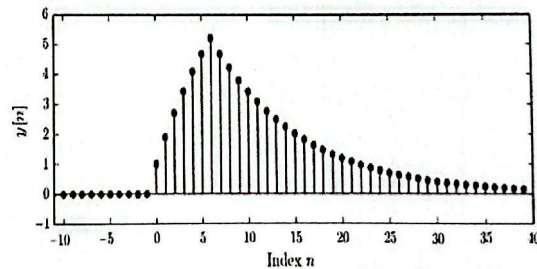
Q1- Signals representation and modeling

- a. [10] The Continuous- time signal $z(t)$ consists of two periodic signals, as shown below. Classify $z(t)$ as either periodic or non-periodic. If it is periodic, determine its time period and fundamental frequency. Furthermore, calculate the number of cycles that two distinct components of the signal will complete within one cycle of $z(t)$

$$z(t) = 5 \cos\left(40\pi t - \frac{\pi}{4}\right) - 3 \sin\left(15\pi t + \frac{3\pi}{2}\right)$$

- b. [10] The Discrete - time signal $y[n]$ is shown below,

$$y[n] = \begin{cases} 0, & n < 0 \\ -9 \left[0.9^n - \frac{1}{0.9} \right], & 0 \leq n \leq 6 \\ 9.8168 (0.9)^n, & n > 6 \end{cases}$$



- Sketch the signal $g[n]$ by performing stepwise necessary operation:

$$g[n] = y[4n + 20]$$

- Calculate the energy of the signal $x[n]$, where $x[n] = g[n] \cdot u[n]$

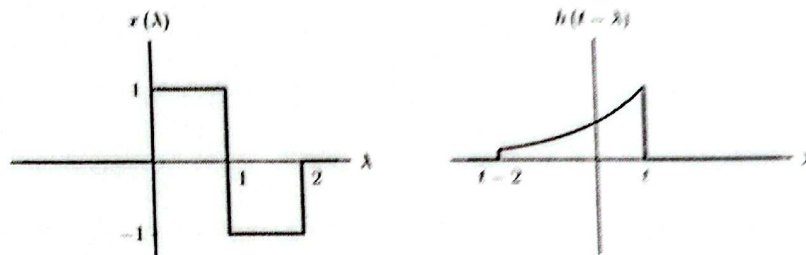
Q2 - Time domain analysis of continuous-time systems

- a. [10] A system model is shown below in terms of its input-output relationship, determine if the given system is linear and/or time-invariant by stepwise application of necessary laws.

$$y(t) = (1 - t) \int_{-\alpha}^t x(\beta) d\beta$$

- b. [10] For the given pair of signals $x(t)$ and $h(t)$, find the convolution $y(t) = x(t) * h(t)$ ONLY for instances $1 < t < 4$. Sketch the signals involved in the convolution integral for each case separately and write the exact expression of $y(t)$ with proper integration limits (the final answer by application of integration operation is not required).

$$h(t - \tau) = e^{-(t-\tau)}$$

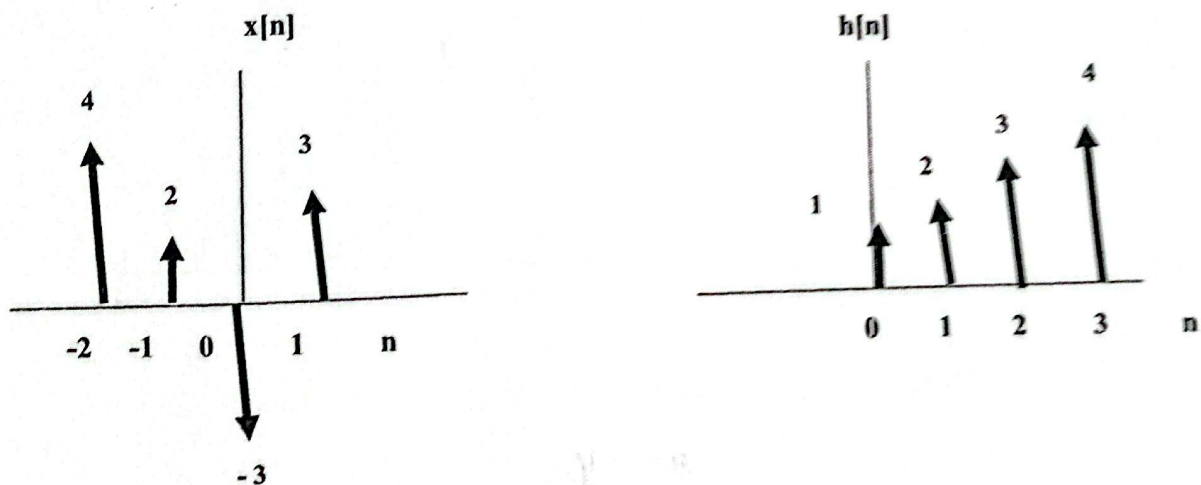


Q3 - Time domain analysis of Discrete-time systems

- a. [10] A system model is shown below in terms of its input-output relationship, determine if the given system is stable and/or causal by stepwise application of necessary conditions.

$$y[n] = \sin\left(-\frac{\pi n}{4}\right) \cdot x[n]$$

- b. [10] For the LTI system, calculate the output $y[n]$ ONLY at $n = 0$ and $n = 3$ by showing the plot and mathematical expression of convolution for both cases separately.



Q. 3) (a) $y[n] = \sin\left(\frac{-\pi n}{4}\right) \cdot x[n]$

causal :

$n < 0$: $y[-1] = \sin\left(\frac{-\pi}{4}\right) \cdot x[-1]$ ✓

$n = 0$: $y[0] = \sin(0) \cdot x[0] = 0$

$n > 0$: $y[1] = \sin\left(\frac{-\pi}{4}\right) \cdot x[1]$ ✓

~~function is causal since it depends only on present value.~~

$y[n]$ depends on current value of $x[n]$
No causal

⇒ function is ~~causal~~ not causal since $y[n] \neq 0$ for all $n < 0$. No

$\sum_{k=-\infty}^{\infty} \sin\left(\frac{-\pi k}{4}\right) = 0 - \frac{\sqrt{2}}{2} - 1 - \frac{\sqrt{2}}{2} + 0 + \frac{\sqrt{2}}{2} + 1 - \frac{\sqrt{2}}{2}$

bounded.

70

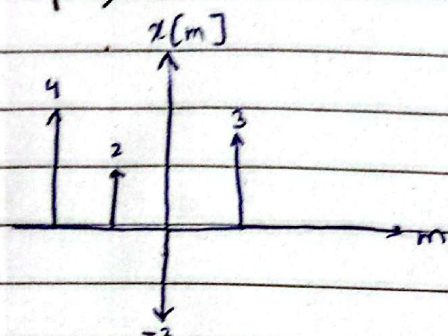
?

is $y[n]$ will be bounded.

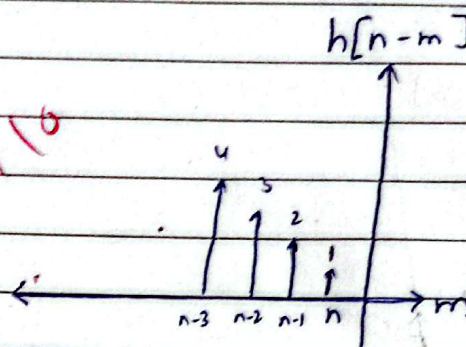
⇒ $\sum_{k=-\infty}^{\infty} y(k) = 0 < 8$ Hence, it is stable.

x 8/10

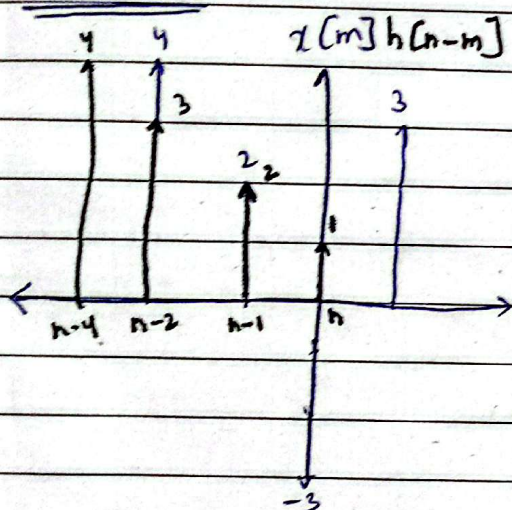
Q.3) (b)



~~10/10~~



at $n=0$:

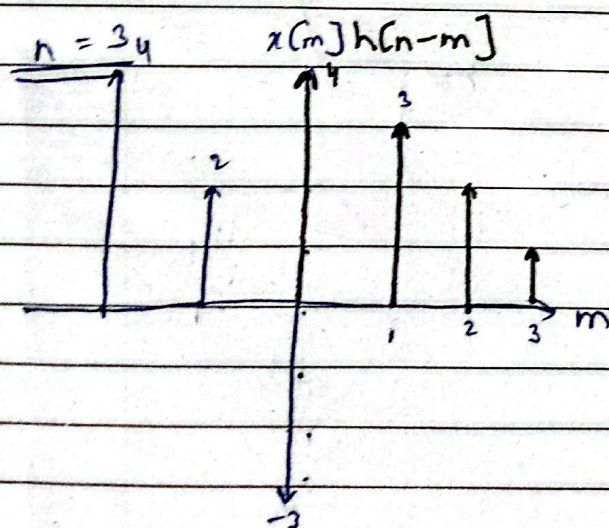


$$y[n] = \sum_{m=-2}^0 x[m]h[n-m]$$

$$y[n] = (-3)(1) + (2)(2) + (4)(3)$$

$$= \cancel{1} + 12$$

$$= 13$$



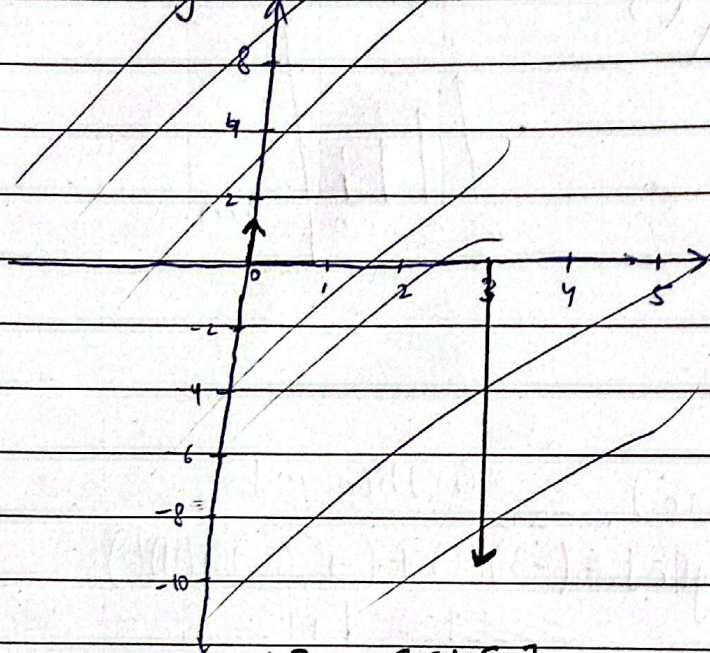
$$y[n] = \sum_{m=0}^1 x[m]h[n-m]$$

$$y[n] = (3)(3) + (-3)(4)$$

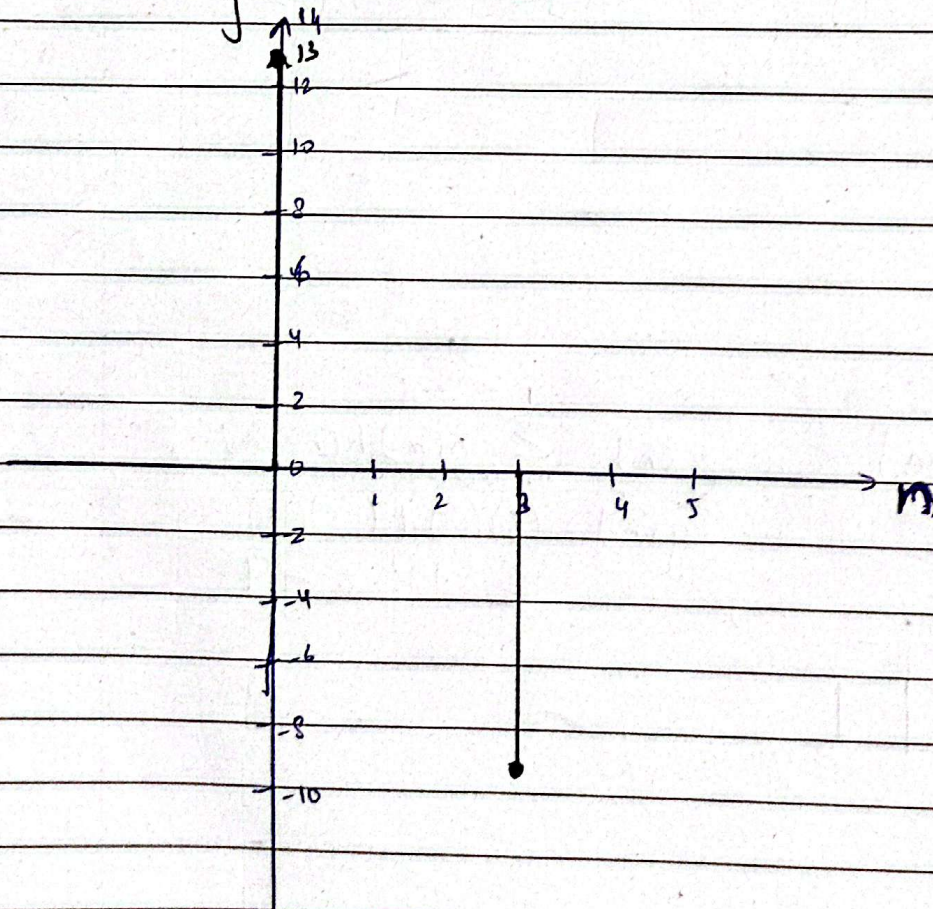
$$= 3 - 12 = -9$$

Output:

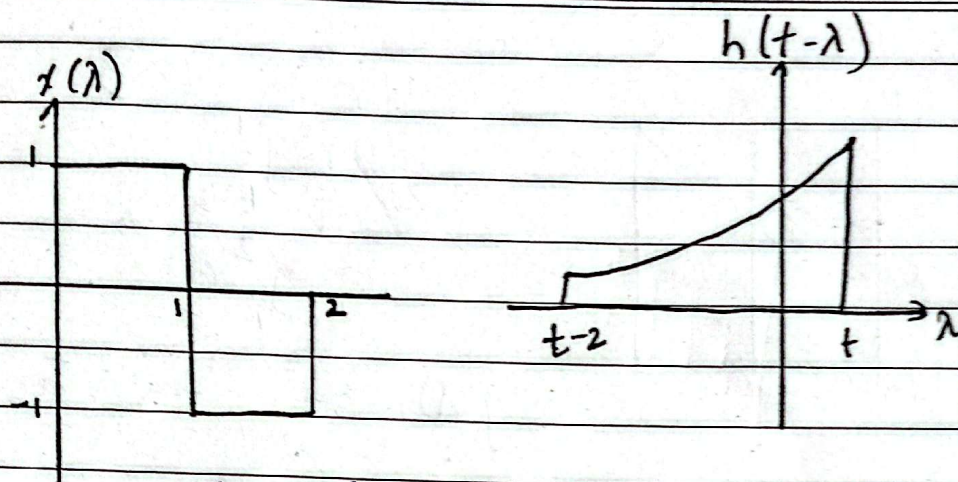
$$y[n] = x[n] * h[n]$$



$$y[n] = x[n] h[n]$$

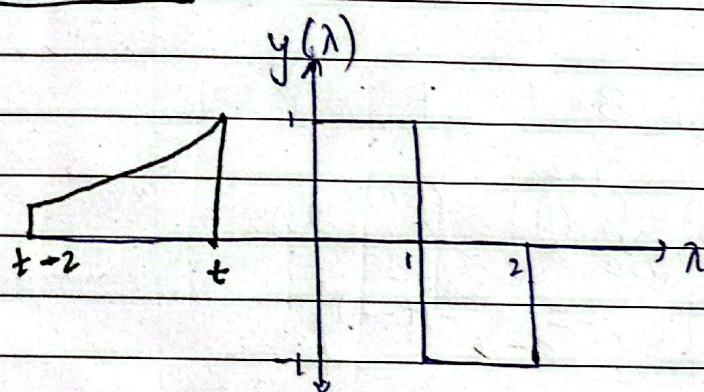


Q.2) (b)



$$y(\lambda) = x(\lambda) * h(t-\lambda)$$

Case #1

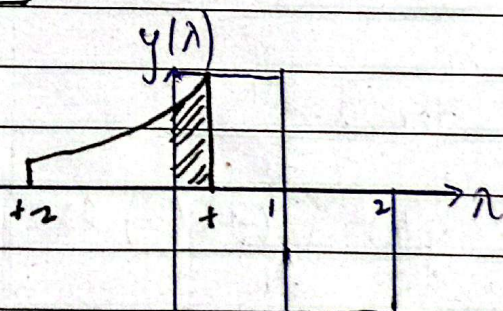


For $t < 0$

$$y(\lambda) = \int_{-\infty}^{\infty} x(\lambda) h(t-\lambda) d\lambda$$

$$= 0$$

Case #2

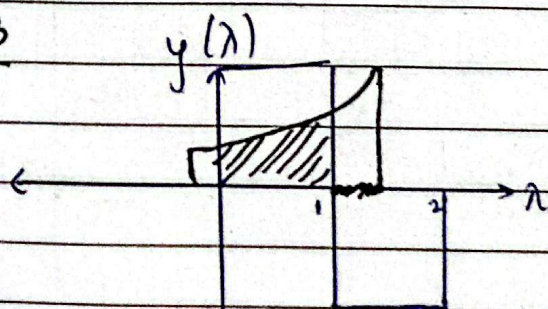


For $0 < t < 1$

$$y(\lambda) = \int_0^t (1) e^{-(t-\lambda)} d\lambda$$

$$= e^{-(t-\lambda)} \Big|_0^t = 1 - e^{-t}$$

Case #3



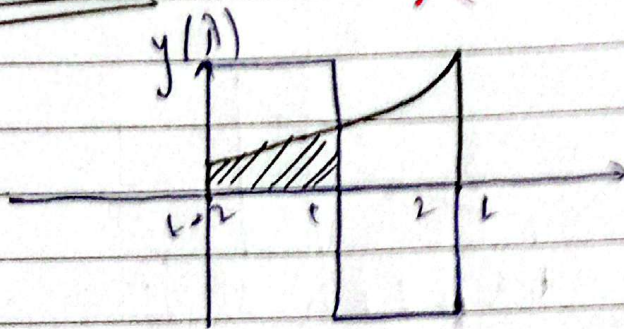
For $1 < t < 2$

$$y(\lambda) = \int_0^1 (1) e^{-(t-\lambda)} d\lambda + \int_1^t (1) e^{-(t-\lambda)} d\lambda$$

$$= e^{-(t-1)} + \left[-e^{-(t-\lambda)} \right]_1^t$$

$$= 2e^{-(t-1)} - 1$$

Case #4:



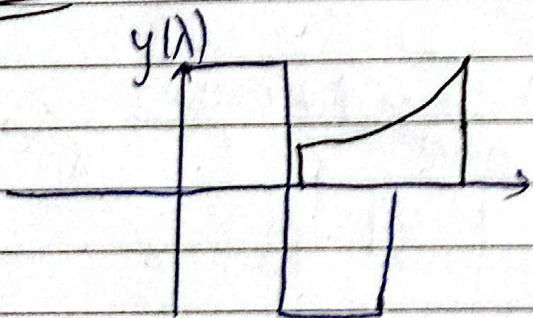
for $2 < t < 3$

$$y(t) = \int_{t-2}^1 (1) e^{-(t-\lambda)} d\lambda + \int_1^2 (\lambda) e^{-(t-\lambda)} d\lambda$$

$$y(t) = e^{-(t-1)} - e^{-2} + e^{-(t-2)} - e^{-2}$$

$$= 2e^{-(t-1)} - e^{-(t-2)} - e^{-2}$$

Case #5:

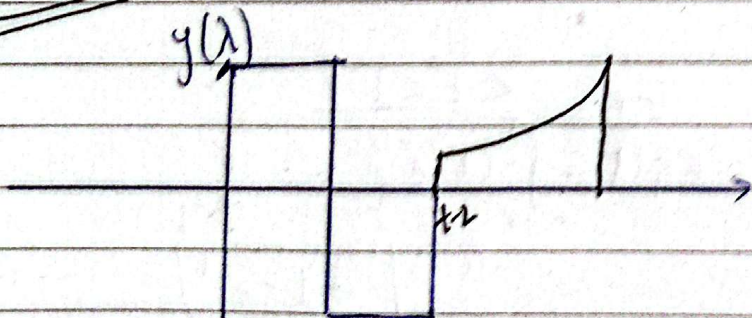


for $3 < t < 4$

$$y(t) = \int_{t-2}^2 (\lambda - 1) e^{-(t-\lambda)} d\lambda$$

$$= e^{-2} - e^{-(t-2)}$$

Case #6:



for $t > 4$

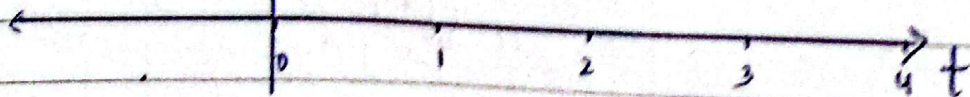
$$y(t) = \int_4^{\infty} (-1)(0) d\lambda$$

$$= 0$$

Output:

$y(t)$

No plotting needed!



$$Q. 2)(a) \quad y(t) = (1-t) \int_{-\infty}^t x(\beta) \cdot d\beta$$

$$y_1(t) = (1-t) \int_{-\infty}^t x_1(\beta) \cdot d\beta \Rightarrow \alpha_1 (1-t) \int_{-\infty}^t x_1(\beta) \cdot d\beta$$

$$y_2(t) = (1-t) \int_{-\infty}^t x_2(\beta) \cdot d\beta \Rightarrow \alpha_2 (1-t) \int_{-\infty}^t x_2(\beta) \cdot d\beta$$

~~$$y(t) = y_1(t) + y_2(t)$$~~

$$y(t) = (1-t) \int_{-\infty}^t [\alpha_1 x_1(\beta) + \alpha_2 x_2(\beta)] \cdot d\beta$$

$$= \alpha_1 (1-t) \int_{-\infty}^t x_1(\beta) \cdot d\beta + \alpha_2 (1-t) \int_{-\infty}^t x_2(\beta) \cdot d\beta$$

$$\Rightarrow y(t) = \alpha_1 y_1(t) + \alpha_2 y_2(t)$$

$$1) \quad x(t) = x(t-t_0) \Rightarrow y(t) = (1-t) \int_{-\infty}^t x(\beta-t_0) \cdot d\beta$$

$$2) \quad y(t) = (1-t) \int_{-\infty}^t x(\beta) \cdot d\beta \Rightarrow y(t-t_0) = (1-t+t_0) \int_{-\infty}^{t-t_0} x(\beta) \cdot d\beta$$

Not same.

* System is Linear But Not Time-Invariant.

Q.1) a) ~~10/10~~
 $x_1(t) = 5 \cos\left(40\pi t - \frac{\pi}{4}\right)$ — (i)

$$40\pi = 2\pi f_1$$

$$f_1 = \frac{40\pi}{2\pi} = 20$$

$$m_1 = 8$$

$$x_2(t) = -3 \sin\left(15\pi t + \frac{3\pi}{2}\right)$$
 — (ii)

$$15\pi = 2\pi f_2$$

$$f_2 = \frac{15\pi}{2\pi} = 7.5$$

$$m_2 = 3$$

$$\Rightarrow \text{fundamental frequency} = \boxed{2.5 \text{ Hz}}$$

$$\Rightarrow \text{Time-period} = \frac{1}{2.5} = \boxed{0.4 \text{ s}}$$

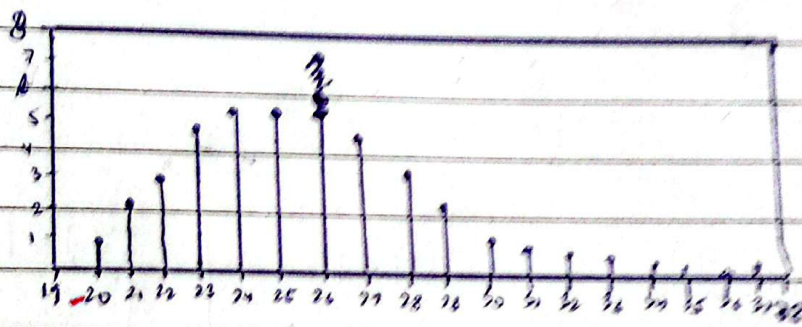
$\Rightarrow$ It is periodic (✓)

$\Rightarrow$ first signal $x_1(t)$ will complete 8 cycles while $x_2(t)$ will complete 3 cycles.

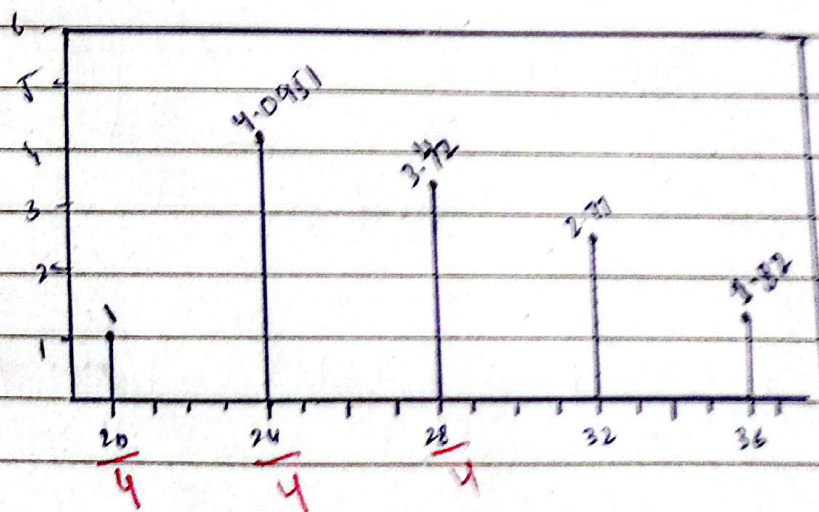
Q.1)
(b)

$$y[n] = \begin{cases} 0 & n < 0 \\ -9 \left[0.9^n - \frac{1}{0.9} \right] & 0 \leq n \leq 6 \\ 9.8168 (0.9)^n & n > 6 \end{cases}$$

$$g[n] = y[4n + 20]$$

 $y[n+20]$
~~shift + 1~~


$$g[n] = y[4n + 20]$$

~~8/10~~


$$\bullet \text{ Calculate energy of } x[n] = g[n] \cdot v[n]$$

$$E_x = \sum_{-\infty}^{\infty} [x[n]^2]$$

$$= (1)^2 + (4.0951)^2 + \sum_{-\infty}^{\infty} 9.8168 (0.9)^{4n+20}$$

$$= 17.77 + 9.8168 (0.9)$$

$$E_x = 26.605$$